

Evidence Synthesis to Inform WHO Road Safety Legal Guidance

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Objectives

To identify effective policy and legislative approaches that support safer roads and reductions in road traffic injury across three priority areas:

1. Distracted driving
2. Speed management and infrastructure
3. Powered two- and three-wheeler (PTW) safety

How can legislation be **designed** to maximise effectiveness?



How can **enforcement** of legislation be improved?



How should legislation consider **equity,** vulnerable users and Safe Systems approaches?



Methods

Scoping review

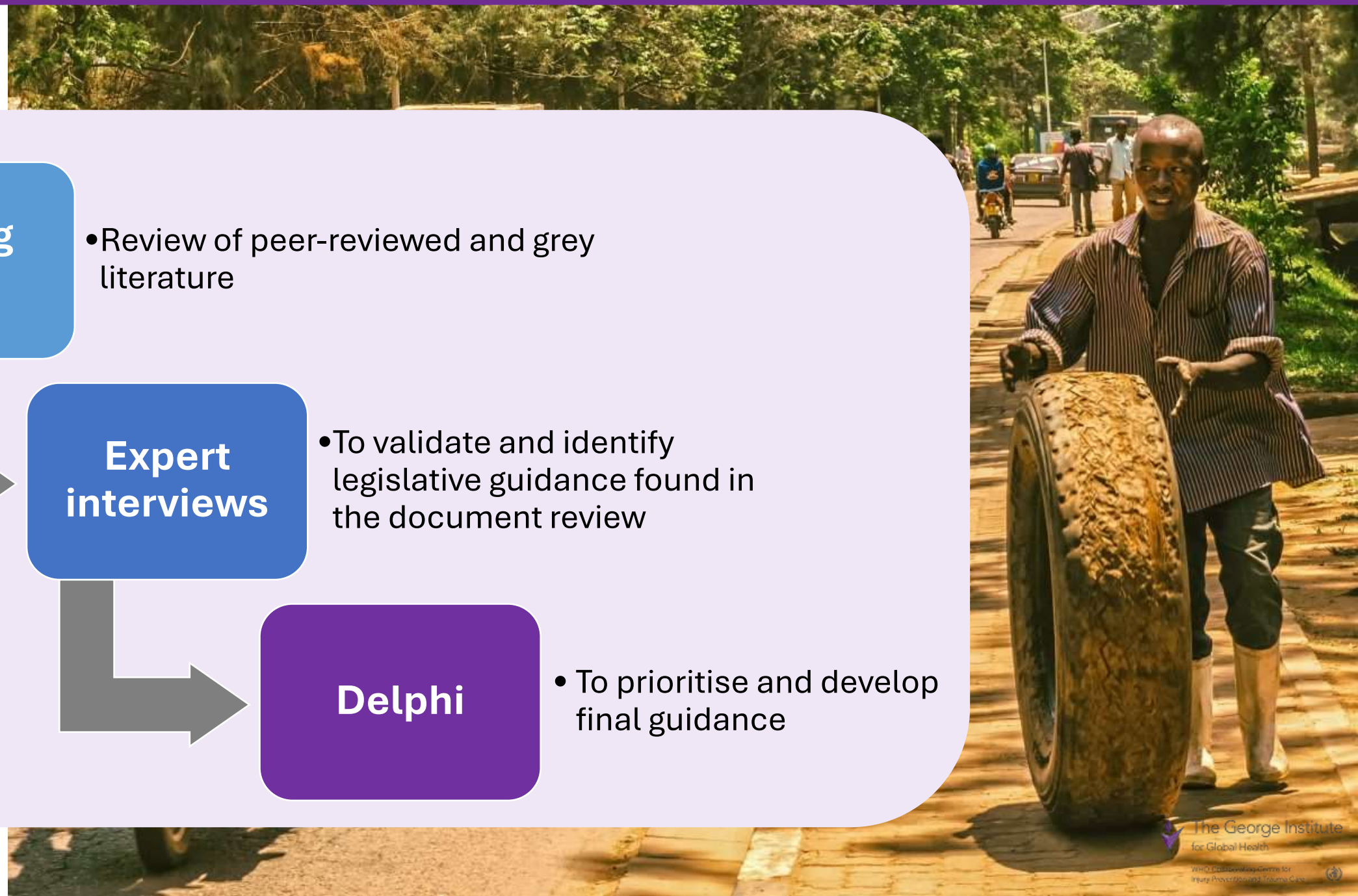
- Review of peer-reviewed and grey literature

Expert interviews

- To validate and identify legislative guidance found in the document review

Delphi

- To prioritise and develop final guidance



Search Strategy

Inclusions

- Legislative, regulatory and policy interventions
- Enforcement, monitoring and evaluation studies
- Peer-reviewed and grey literature
- Road safety outcomes and implementation evidence from policy/legislation

Exclusion

- Behavioural/Interventional studies without policy relevance or significant discussion
- Non-road-safety interventions
- Editorials, opinion pieces and abstracts



Search strategy

Component	Peer-Reviewed Literature	Grey Literature
Years searched	2005-2026	
Languages searched	English, Arabic, Spanish/Portuguese, Chinese	
English	MEDLINE/PubMed; Embase; Web of Science; PsycINFO; TRID	WHO IRIS; ITF/OECD; World Bank; iRAP; SWOV; TRL; ETSC
Arabic	Dar Almandumah, Journal websites, Others	Qatar University QSpace
Spanish / Portuguese	SciElo, LILACS/VHT , Redalyc, Latindex, Dialnet	PAHO IRIS, ITF/OECD, TESEO, World Bank
Mandarin	China National Knowledge Infrastructure, Wanfang Data	People's Public Security Daily
Publication	2005-2026	2005-2026
Quality appraisal	MMAT	AACODS

Distracted Driving



Overview of findings

91
SOURCES INCLUDED

59
PEER-REVIEWED STUDIES

32
GREY LITERATURE DOCUMENTS

2005—2026
PUBLICATION WINDOW

MIXED EVIDENCE
Evaluations • mixed methods • legislative analysis • implementation guidance

WHERE THE EVIDENCE COMES FROM

48 United States

10 Europe/EU
7 Australia
6 China
3 Canada

HIGH-INCOME SKEW
LMIC-specific studies were scarce; African evidence came mainly from multi-country and international reports

WHAT THE LITERATURE EXAMINES

Mobile-phone legislation

Handheld bans • texting bans • wider cellphone restrictions

Novice-driver rules • penalties • offence classifications

Peer-reviewed evidence was mainly quantitative, non-randomised evaluation

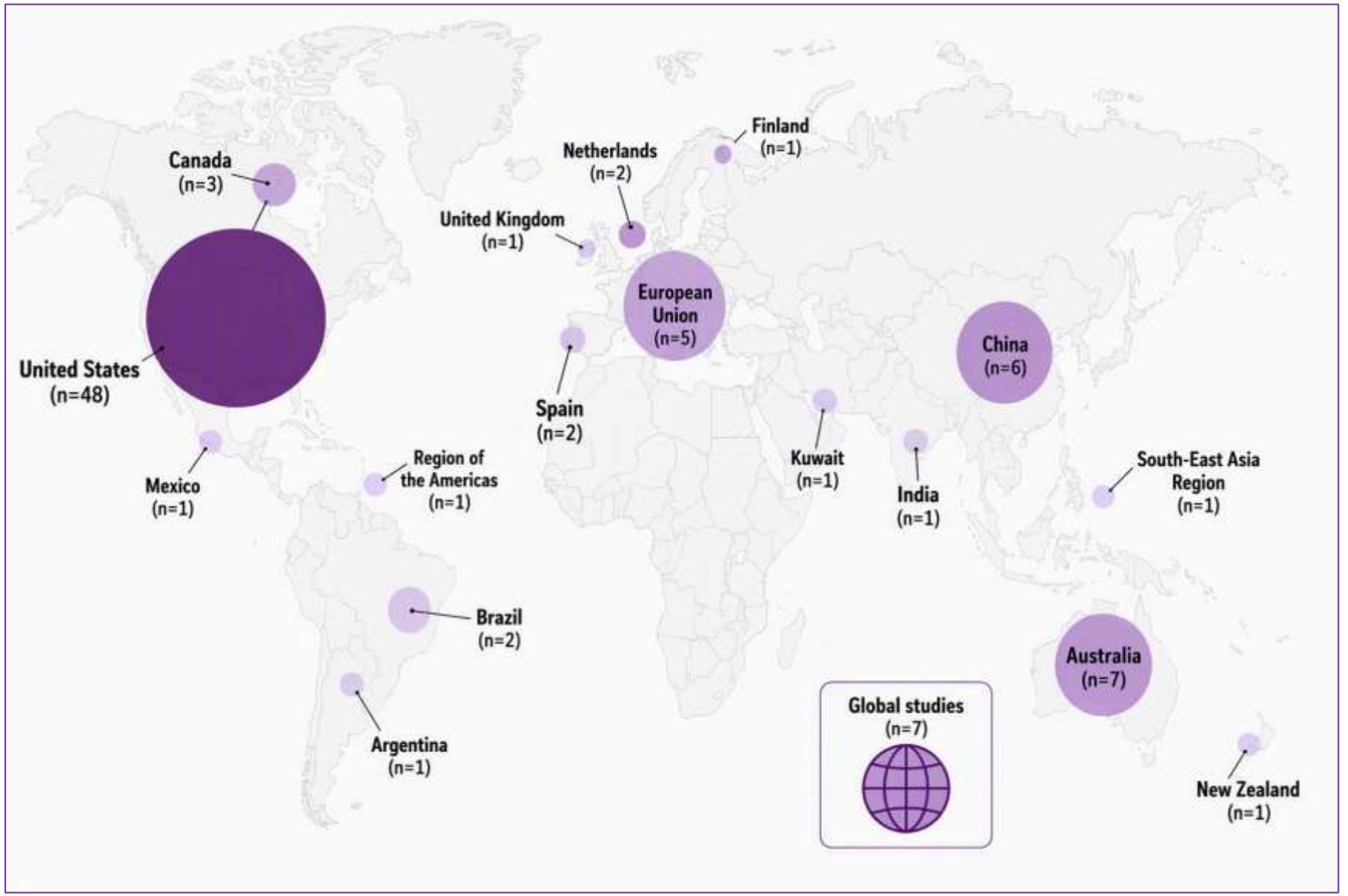
HOW LAWS WORK IN PRACTICE

Enforcement & compliance most discussed

Followed by implementation, monitoring & evaluation

LEAST DEVELOPED
Legislature development, governance & institutional roles, employer policy, vehicle standards and legislative review.

Distracted Driver Evidence Landscape



QUESTION 1: What issues should be considered if distracted driving legislation is to be effective?

Effective laws are clear, universal and enforceable

01

Clear definitions

Use standardised, technology-neutral language that covers broad device handling and not just texting or calls.

CLOSE LOOPHOLES

02

Universal coverage

Apply the law to all drivers. Keep added safeguards for novice drivers without creating a fragmented rule set.

ONE CLEAR STANDARD

03

Primary offence

Allow police to intervene for distracted driving itself, without first observing another traffic violation.

MAKE IT ENFORCEABLE

QUESTION 1: What issues should be considered if distracted driving legislation is to be effective?

Effective laws pair credible sanctions with support and renewal

04

Proportionate penalties

Use graduated sanctions and licensing measures. Higher fines alone are weak when detection is not credible.

ESCALATE FAIRLY

05

Education & norms

Pair the law with sustained education and publicity so expectations are understood and socially reinforced.

BUILD COMPLIANCE

06

Built-in review

Schedule evidence-led updates as vehicle interfaces, technologies and driver behaviours continue to change.

STAY CURRENT

QUESTION 2: What factors should be considered to ensure effective enforcement?

Detection works when it is targeted, visible and understood

01

Targeted detection

Use high-vantage patrols, spotters, unmarked vehicles and technology where risks and non-compliance are greatest.

DETECT RELIABLY

02

Sustained visibility

Maintain regular, highly visible operations so drivers continue to believe violations are likely to be detected.

KEEP RISK VISIBLE

03

Public communication

Publicise enforcement activity, warning periods and penalties, amplify roadside presence.

BUILD AWARENESS

QUESTION 2: What factors should be considered to ensure effective enforcement?

Operational systems determine whether the law has real reach

04

Capacity & training

Provide dedicated staff, funding, equipment and ongoing officer training beyond routine policing

RESOURCE THE WORK

05

System coordination

Align police, prosecutors, courts, transport and road-safety agencies from detection through sanction.

CONNECT THE CHAIN

06

Performance monitoring

Track officer activity, citations, awareness, compliance and outcomes using shared definitions and data.

LEARN & IMPROVE

QUESTION 3: What factors should be considered for monitoring and review of law?

Monitoring must be comparable, connected and ready for change

01

Standardise data

Use common definitions, coding and reporting methods across places and over time.

COMPARE CONSISTENTLY

02

Integrate sources

Link police, health, insurance, licensing and observational data to reduce blind spots.

COMPLETE THE PICTURE

03

Set indicators

Define performance measures, targets and benchmarks that show whether the law is working.

TRACK PROGRESS

04

Measure behaviour

Monitor phone use and compliance in addition to crashes, which often underreport distraction.

SEE CHANGE EARLY

05

Scan new risks

Update categories and surveillance as apps, wearables and in-vehicle systems evolve.

STAY AHEAD

Equity and Safe System coverage remains thin

3 Sources discussed equity or vulnerability

21% engaged with Safe System thinking

Guidance has moved to shared responsibility; legislation mostly has not.

EQUITY

Impacts are rarely tested

Fixed fines, demerit points and license loss may fall hardest on lower-income, young and novice drivers, but differential impacts are seldom evaluated.

VULNERABLE ROAD USERS

Coverage is still car-centred

Cyclists, motorcyclists and informal transport users face distraction risk, but most laws and evaluations focus on passenger-vehicle drivers.

SAFE SYSTEM

Endorsed more than embedded

Guidance promotes safer vehicles, interfaces, monitoring and shared responsibility, but few laws explicitly assign duties beyond drivers and police.

IMPLICATION **Test distributional effects and shift from driver-only rules to proportionate penalties, shared responsibility and safer system design.**

Distracted Driver Key Policy Recommendations

A practical policy route: scope it, enforce it, measure it, share it, make it fair.

01

Future-proof the law

Cover all populations; Cover current and emerging technology with clear, technology-neutral rules.

02

Build enforcement in

Pair practical powers with trained officers, resources, detection technology and sustained communication.

03

Make data do the work

Collect and link standardised behavioural, enforcement and crash indicators, and use them to improve policy.

04

Share the steering

Assign a lead agency, and assign clear roles, engage employers, manufacturers, and technology providers.

05

Design for everyone

Test equity impacts; include vulnerable road users, LMIC conditions and Safe System principles.

Design the whole system to prevent harm

Speed and *"infrastructure management"*



Speed-law evidence is broad-but unevenly distributed

76
SOURCES INCLUDED

48
EMPIRICAL STUDIES

28
GREY-LITERATURE ITEMS

2006—2026
PUBLICATION WINDOW

MIXED EVIDENCE
26 non-empirical sources • case studies • commentary • reviews

WHERE THE EVIDENCE COMES FROM

17 China

14 United States
6 Australia
6 Canada
6 United Kingdom

GLOBAL BUT UNEVEN
61 single-country studies across 17 countries; 15 were multi-country. Africa appeared in only two studies.

WHAT THE LITERATURE EXAMINES

Speed laws & enforcement

88% addressed speed limits and conduct rules

76% addressed enforcement and compliance

66% focused on speed alone; **25%** treated speed within broader road-safety law.

WHAT REMAINS UNDERDEVELOPED

Safe System & equity

Vulnerability and equity appeared in **29%** of studies; Safe System principles in **20%.**

EVIDENCE QUALITY
Much of the literature was descriptive or non-empirical rather than rigorous evaluation.

Speed Evidence Landscape

Geographic distribution of included studies

n = 76 studies; 19 countries represented



A further 15 studies are supranational or multi-country (global, EU/Europe, OECD) and are not mapped. Subnational studies are attributed to their national jurisdiction.

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WHO Collaborating Centre for Injury Prevention and Trauma Care

QUESTION 1: What makes speed-management legislation effective?

Safe limits must reflect people, place and road risk

01

Safe limits

Use ≤50 km/h in urban areas and 30 km/h where vehicles mix with pedestrians, cyclists or other unprotected users.

PROTECT PEOPLE

02

Evidence-led setting

Set and revise limits using road function, traffic mix, conflict risk and human injury tolerance.

SET SPEEDS SAFELY

03

Roads reinforce limits

Align posted limits with road protection; use self-explaining, forgiving design and safety audits to make compliance credible.

DESIGN FOR COMPLIANCE

QUESTION 1: What makes speed-management legislation effective?

Authority, sanctions and review turn limits into action

04

Enforcement-ready law

Authorise enforcement methods and define evidence, device approval, calibration, certification, liability and notification rules.

MAKE IT OPERABLE

05

Credible sanctions

Graduate penalties by severity; combine fines, points and licence action, with stronger responses for extreme or repeat speeding.

DETER CONSISTENTLY

06

Govern & review

Assign funded duties across agencies and require periodic review triggered by road function, traffic, crashes and user vulnerability.

KEEP POLICY CURRENT

Source: Speed Legislation Scoping Review — Results, section 3

QUESTION 2: What shapes effective speed legislation, enforcement and review?

Effective systems need authority, legitimacy and delivery capacity

01

Authority that can act

Place power where agencies can respond, and ensure enabling law covers devices, evidence, liability and due process.

REMOVE LEGAL BLOCKS

02

Political & public legitimacy

Prepare for policy windows; use a clear safety purpose, consultation, transparent siting and accountable revenue use.

BUILD TRUST

03

Capacity & governance

Assign funded duties, a strong lead agency, cross-sector coordination and sufficient local technical capability.

ENABLE DELIVERY

Source: Speed Legislation Scoping Review — Results, Table 2

QUESTION 2: What shapes effective speed legislation, enforcement and review?

The chain succeeds only when detection leads to learning and reform

04

Sustained enforcement

Use frequent activity, mixed visible and covert methods, evidence-led sites and low tolerances; detection certainty matters most.

DETER CONSISTENTLY

05

Complete admin chain

Connect certified devices, reliable records, prompt notice, scalable processing, adjudication and payment.

CLOSE EVERY GAP

06

Data-to-law feedback

Track speeds, crashes and casualties; mandate audits, review triggers, public reporting and revision of the law.

LEARN & ADAPT

Source: Speed Legislation Scoping Review — Results, Table 2

Equity and Safe System coverage remains thin

29% Addressed vulnerability or equity

20% Engaged with Safe System thinking

Both are usually used as rationale, but not translated into legal duties.

EQUITY

Rarely operationalised

Enforcement location, fine levels and socioeconomic impacts are seldom assessed; officer-led enforcement may create disproportionate risks.

VULNERABLE ROAD USERS

Used to justify lower speeds

Pedestrians and cyclists underpin 30 km/h recommendations, but are rarely given explicit rights, protections or priority in law.

SAFE SYSTEM

Principles stop short of law

Human tolerance and shared responsibility guide limits and design, yet mostly appear in technical guidance rather than binding duties.

IMPLICATION **Embed equity checks, explicit vulnerable-user duties and shared system responsibility directly in the legal framework.**

Source: Speed Legislation Scoping Review — Results, §5

Speed Management and Infrastructure Key Policy Recommendations

A practical policy route: set safe speeds, assign duties, enforce, learn, protect.

01

Set safe speeds

Set and revise limits by road function, traffic mix, conflict risk and human tolerance.

02

Enable agencies

Clarify authority and duties; fund technical capacity and administration.

03

Build in enforcement

Authorise methods; define evidence, device standards, liability, sanctions and appeals.

04

Monitor and adapt

Audit risky speeds and serious harm, then use findings to revise policy.

05

Protect everyone

Embed equity and vulnerable-user duties in enforcement, fines and safe-infrastructure investment.

Design law, roads and enforcement as one Safe System

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Powered two- and three-wheelers



Why PTW safety matters more than ever

Essential for daily life

PTWs provide affordable, flexible transport and connect people to employment, education, health care and markets.

Rapid and sustained growth

The global PTW fleet has grown dramatically, driven by urbanisation, gaps in public transport and increasing demand for ride hailing and delivery services.

Widely used in low- and middle-income countries

Make up a large share of the total vehicle fleet in many countries and are a major part of the transport system.

The policy challenge: mobility is evolving faster than regulation

Newer forms of mobility, including electric motorcycles, e-bikes and e-scooters and commercial platform based services are expanding the system and creating new regulatory challenges



≈500 million

motorcycles registered globally



Of all registered vehicles worldwide



≈1 in 3

Road traffic deaths are motorcyclists

The George Institute for Global Health

WHO Collaborating Centre for Injury Prevention and Trauma Care

Overview of the PTW evidence base

144
SOURCES INCLUDED

73
EMPIRICAL STUDIES

45
GREY-LITERATURE ITEMS

2006—2026
PUBLICATION WINDOW

MIXED EVIDENCE
26 non-empirical sources • case studies • commentary • reviews

WHERE THE EVIDENCE COMES FROM

42 China

31 United States
11 Viet Nam
6 Spain
5 Iran

GLOBAL BUT UNEVEN
Evidence from 37 countries across all 6 WHO regions.
Africa and south America were underrepresented

WHAT THE LITERATURE EXAMINES

Evidence concentrated on helmet legislation

The strongest evaluated evidence concerned stand-alone helmet-use legislation, particularly helmet use and injury outcomes.

Grey literature broadened the evidence to vehicle regulation, registration, e-mobility, commercial operation, enforcement, governance and lifecycle control.

26 further sources provided legal, policy, historical and review evidence.

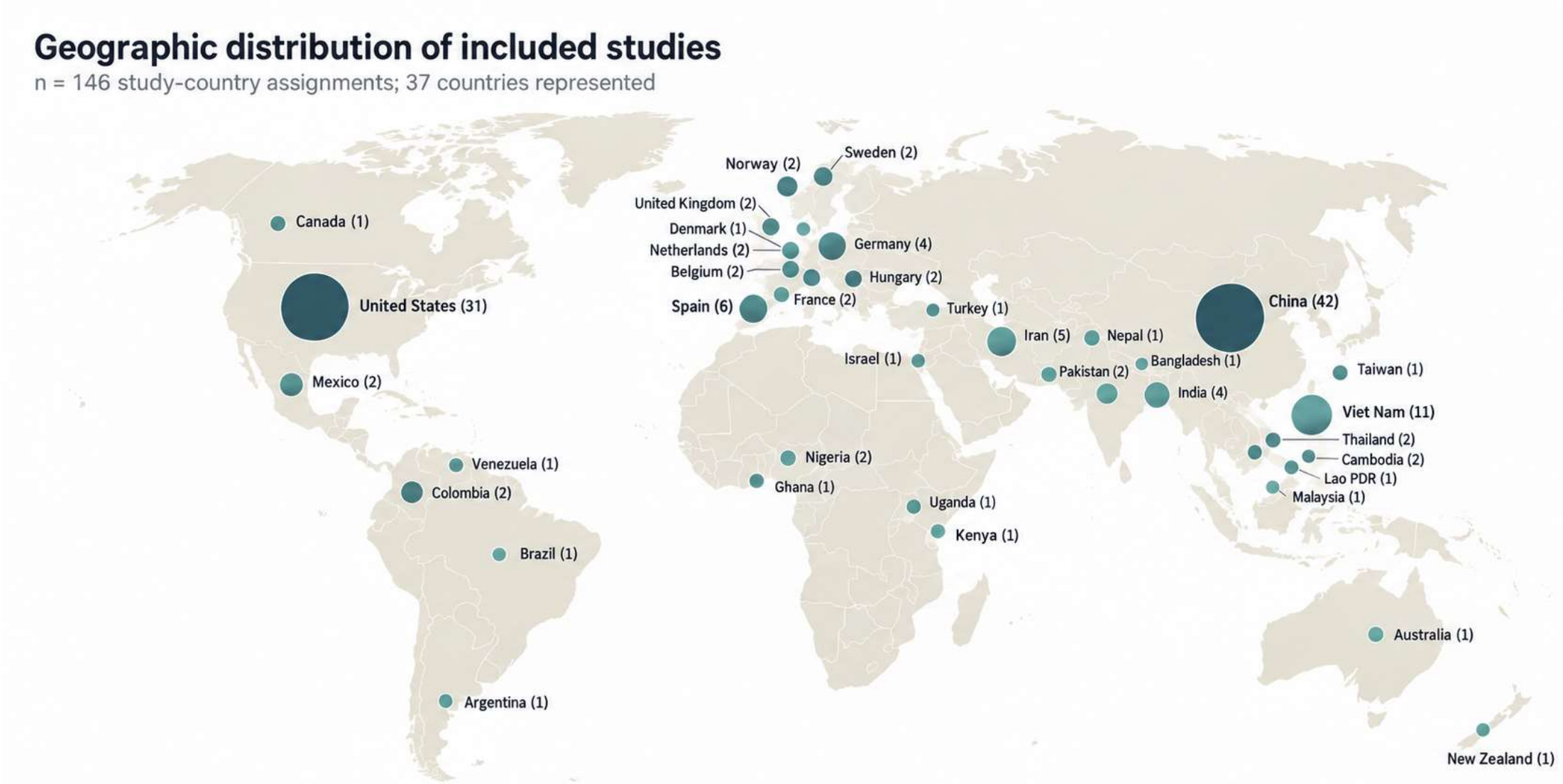
WHAT REMAINS UNDERDEVELOPED

Whole-system evidence remains limited

Direct evidence remained limited for infrastructure and speed, vehicle regulation, commercial and platform responsibilities, post-crash provisions, governance and integration with the wider transport system.

Evidence on three-wheelers was limited. Equity and Safe System principles were also inconsistently examined.

Powered Two- and Three-Wheeler Evidence Landscape



Counts reflect study-country assignments. Multi-country studies were counted in each named country.
Broad global or supranational studies were not mapped.

George Institute

QUESTION 1: What makes powered two- and three-wheeler legislation effective?

The legislation needs to be comprehensive and fit for a changing mobility system

01

Clear scope & coverage

Clearly define the vehicles, users, roads and activities covered. Minimise exemptions and avoid regulatory gaps as new PTW types and uses emerge.

DEFINE WHO & WHAT IS COVERED

02

Shared responsibility

Do not place responsibility solely on riders. Allocate duties to government, road authorities, regulators, manufacturers, sellers, repairers, commercial operators and platforms.

SHARE RESPONSIBILITY

03

Comprehensive safety provisions

Address risks across road users, vehicles, infrastructure and speed, commercial operation and post-crash response, rather than relying mainly on rider behaviour.

ADDRESS THE WHOLE SYSTEM

QUESTION 1: What makes powered two- and three-wheeler legislation effective?

Legislation must translate broad safety goals into clear requirements

04

Specific requirements

Set clear requirements for helmet standards, licensing, minimum age, passengers and loads, registration, vehicle standards, insurance and commercial operation.

05

Risk-matched regulation

Requirements should reflect vehicle power, speed, use and risk. This includes graduated licensing/classification and rules appropriate to different PTW types and uses.

06

Adaptable legislation

Definitions and requirements need to keep pace with e-bikes, electric motorcycles, modification, delivery/platform use and other changing mobility patterns.

“MOBILITY IS CHANGING FASTER THAN ROAD SAFETY POLICY”

QUESTION 2: What shapes implementation, enforcement & review of PTW laws?

Legislation only protects people when it can work in practice

01

Implementable

Registration, licensing and compliance systems must be accessible and workable in practice.

MAKE COMPLIANCE POSSIBLE

02

Enforceable

Responsibilities must be clear, enforcement agencies adequately resourced and sanctions practical, proportionate and consistently applied.

MAKE THE LAW ENFORCABLE

03

Coordinated

Government, road authorities, police, licensing bodies, vehicle regulators and local authorities need clearly defined and coordinated responsibilities.

DEFINE WHO DOES WHAT

QUESTION 2: What shapes effective implementation and enforcement?

Effective regulation must learn and adapt

01

Monitor

Track implementation, compliance, enforcement, crashes and injuries to determine whether legislation is working as intended.

02

Evaluate

Assess effectiveness and unintended effects rather than assuming that enactment alone will produce safer outcomes.

03

Review & adapt

Use monitoring and evaluation to revise requirements, address implementation problems and respond to changes in vehicles, technology and mobility.

QUESTION 3: How is PTW safety addressed across the main regulatory domains?

The evidence extends well beyond helmet-use legislation

01

Road users

Helmet use and standards, licensing, minimum age, training, impairment, passengers, loads and other rider requirements.

02

Vehicles

Vehicle standards, registration, roadworthiness, modification, identification and requirements for emerging and electric PTWs.

03

Roads & speeds

Speed management, infrastructure design, road-space allocation and the interaction of PTWs with other road users.

QUESTION 3: How is PTW safety addressed across the main regulatory domains?

A comprehensive framework also reaches beyond the rider, vehicle and road

01

Commercial operation

Regulate commercial and delivery use, including operator and platform responsibilities, working practices, vehicle safety, insurance and compliance.

02

Governance & post-crash

Define institutional responsibilities, accountability, data and monitoring requirements, insurance and relevant post-crash duties.

03

Transport integration

Consider PTWs within the wider mobility system, including their relationship with public transport, alternative modes, road-space allocation and changing travel patterns.

QUESTION 4: What does the evidence tell us about gaps and priorities?

The evidence base is strongest for rider-focused legislation and thinner across the wider system

01

Established evidence

The most developed evidence concerns stand-alone helmet-use legislation, particularly helmet wearing and injury outcomes.

02

Major evidence gaps

Direct evaluation is much more limited for comprehensive PTW frameworks, infrastructure duties, vehicle regulation, commercial responsibilities, governance, post-crash provisions and transport integration.

03

Broader direction

Across the evidence, PTW safety increasingly points towards combining individual road-user duties with responsibilities across vehicles, roads, institutions and the wider mobility system.

Equity & Safe System principles are inconsistently translated into PTW regulation

EQUITY

Unequal effects are often overlooked

Costs of helmets, licensing, registration, insurance and vehicle compliance can fall disproportionately on lower-income users and those dependent on PTWs for mobility or livelihoods.

VULNERABLE ROAD USERS

Protection is not always equal

Passengers, children and other groups may receive different protection through exemptions, differential requirements or uneven enforcement.

SAFE SYSTEM

Responsibility still centres on the rider

Much of the evidence focuses on helmet use, licensing and rider behaviour, while explicit duties for road authorities, regulators, manufacturers, commercial operators and other system actors are less developed.

IMPLICATION **Embed equity assessment and shared responsibility across the legal framework, rather than treating them as separate considerations.**

Powered Two- and Three-Wheeler Key Policy Considerations

A practical legal framework: define, protect, share responsibility, implement, adapt.

01

Define the system

Clearly define PTW types, users and uses, with classifications that can accommodate emerging vehicles and changing mobility.

02

Protect across the Safe System

Clarify authority and duties; fund technical capacity and administration.

03

Share responsibility

Set duties for riders and passengers, but also for government, road authorities, regulators, manufacturers, sellers, commercial operators and platforms.

04

Make legislation work

Build in practical registration and licensing, institutional capacity, enforcement powers, coordination and accessible routes to compliance.

05

Monitor, adapt and protect equity

Use compliance, crash and injury data to review legislation as mobility changes, while considering affordability, access and unequal impacts.

Move from stand-alone rider requirements towards a comprehensive, shared-responsibility Safe System framework

Conclusions



Five themes recur across all three legislative reviews

01

Write for change

Use clear, comprehensive rules that adapt to populations, technology, vehicle and context.

02

Make law implementable

Fund training, administration and infrastructure, with realistic compliance for people and agencies.

03

Enforce credibly

Define powers and proportionate sanctions; sustain visible enforcement, public communication and detection.

04

Share responsibility

Assign a lead agency and duties across government, industry, employers, operators and road authorities.

05

Monitor, learn and protect

Link data to review; test effects on vulnerable groups, equity, mobility and livelihoods.



The evidence base has five recurring blind spots

Across all three reviews, research describes the legislation better than it tests the system.

1 Causality is unclear	Bundled reforms and observational designs rarely isolate the law from enforcement, publicity, infrastructure or exposure.
2 Implementation is unclear	Few studies follow how laws are developed, resourced, coordinated and translated into everyday practice.
3 The system beyond users is under-tested	Infrastructure, vehicles, technology, employers, platforms, governance and post-crash duties lack rigorous evaluation.
4 Equity is seldom measured	Distributional effects on vulnerable users, lower-income groups, workers, mobility and livelihoods remain poorly quantified.
5 Coverage does not match the risk	LMICs, three-wheelers, informal and commercial transport, emerging technology and long-term monitoring are underrepresented.

Thank you

Arabic

Sara Isameldin Mohamed & Rania Hamada Saad Abdelhamid, World Health Organization

Mandarin

Zhou Yanru, Tongji University

Spanish/Portuguese

Alejandra Piragauta, The George Institute for Global Health, UK

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WHO Collaborating Centre for Injury Prevention and Trauma Care